

A Low-Cost Single-Camera Adaptive Signal Controller for Heterogeneous Urban Traffic: Queue-Weighted Pressure Control with Bounded Fairness and Emergency Preemption

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Abstract

Fixed-time signal plans remain the default in many South Asian cities, where traffic is motorcycle-dominated, demand is strongly asymmetric between approaches, and the budget for inductive loops or radar is unavailable. We present R.E.L.A.Y., an adaptive signal controller whose only input is one existing fixed junction camera. A single convolutional detector produces per-approach, per-class vehicle counts, which are converted to passenger-car-unit (PCU) demand and consumed by a queue-weighted pressure controller in the max-pressure family. The controller adds the operational rules a deployable signal needs and a plain argmax policy lacks: empty-phase skipping, gap-out, minimum and maximum green, a hard anti-starvation bound, pedestrian demand treated as pressure rather than as a fixed interval, and emergency preemption with a bounded hold. We evaluate it in a seeded point-queue microsimulation against an equal-split fixed-time plan and against a deliberately favourable reference whose cycle and splits come from Webster’s formulae applied to the true mean demand. Across 30 seeds per cell and five demand levels, mean delay falls by 14.2% to 38.8% against the Webster plan under asymmetric demand and by 22.1% to 37.6% when a single approach dominates, with the 95th-percentile wait cut from 25.8 s to 15.8 s at 1278 veh/h. We also report two negative results. Under balanced saturated demand the controller is 30.5% worse than the Webster plan, because unconstrained switching raises the phase change rate from 204 to 359 per hour and cuts displayed green time from 0.743 to 0.550 of wall time. In a three-junction corridor, disabling the downstream coupling term changes mean delay by less than 2.5%, so the corridor gain comes from local adaptivity rather than coordination. Emergency preemption reaches green in 4.19 s on average over 900 trials, with a measured worst case of 13.85 s against an analytical bound of 14.0 s. The perception stack runs at a median of 38.3 ms per frame at 960 px inference on a consumer laptop, but on motorcycle-dense footage it undercounts severely, with sampled counts rising 83% between 640 px and 1600 px without converging. This is a simulation and real-footage feasibility study, not a field deployment.

Keywords: adaptive traffic signal control; max-pressure control; computer vision; heterogeneous traffic; emergency vehicle preemption; low-cost intelligent transport systems

1 Introduction

Urban signal control in low- and middle-income cities is usually performed by fixed-time plans or by police officers operating signals manually. Both allocate green without measuring what is queued. The consequence is visible at almost any asymmetric junction: a busy arterial approach holds a long queue while the crossing minor street receives the green it would receive at three in the

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morning. Webster’s method [1] showed that cycle length and splits should follow demand, but a fixed plan can only follow the demand measured when the plan was designed, and re-timing surveys are infrequent. Studies of signalised intersections in the Kathmandu valley report saturation flows and delays that differ substantially from standard capacity guidance, attributing the difference to the local vehicle mixture and to lane discipline [2, 3], which is exactly the condition under which a plan imported from a design manual performs badly.

The established remedy is a demand-responsive system such as SCOOT [4, 5] or SCATS [6], or an optimisation-based controller in the OPAC [7] and RHODES [8] lineage. These work and have decades of field evidence behind them, but they are expensive: they assume a detector plant, communications and a central control room, so per-junction cost is dominated by sensing and civil works rather than by the algorithm. In a city where the signal head itself is sometimes unpowered, that capital profile is prohibitive.

Two conditions have changed. Fixed cameras are already installed at many urban junctions for enforcement and monitoring, so a camera-based controller reuses a paid-for asset. And single-stage object detectors are now accurate and cheap enough to run at video rate on commodity hardware [9, 10], which makes per-approach vehicle counting a software problem rather than a sensor-procurement problem. Vision-based queue and density estimation from fixed cameras is an established technique in its own right [11–13], and several recent studies connect a detector to a signal timer on this basis [14–17].

What is often missing in that applied literature is the control side. Many camera-based studies compute a vehicle count and set green time in proportion to it, which is a proportional rule rather than a control policy, and they rarely state what prevents a minor approach from being starved, what happens when the detector fails, or how the policy behaves at saturation. Conversely, the theoretical literature on max-pressure control [18, 19] provides throughput-optimality results in a form that assumes ideal queue observation and imposes no minimum green, no clearance interval and no fairness bound, all of which a deployable signal requires.

This paper describes and evaluates R.E.L.A.Y., a controller placed deliberately between those positions. Its contributions are: a single-camera perception-to-control pipeline for heterogeneous traffic in which presence and PCU-weighted demand are kept as separate quantities, because a PCU-weighted score alone classifies a lone motorcycle as an empty lane; a queue-weighted pressure controller whose fairness is enforced by separate maximum-wait bounds for vehicles and pedestrians and by a cap on the preemption hold rather than left to the score; an evaluation against a Webster plan computed from the true mean demand, much harder to beat than the equal-split baseline camera-based studies usually use, with an explicit report of where our controller loses and why; and a measured preemption latency distribution compared against a closed-form worst-case bound.

We are explicit about what this is not. There is no field installation, no controller cabinet in the loop and no before-and-after study on a real signal: control results come from a seeded microsimulation and perception results from recorded footage. The contribution is a feasibility and design study at the level a city could use to decide whether a pilot is worth funding, and Section 7 states which claims a field trial would still have to establish.

2 Related work

2.1 Fixed-time and demand-responsive control

Webster’s analysis of cycle length and green splits [1] remains the reference method for fixed-time plans and is the basis of the stronger baseline used here. Among demand-responsive systems, SCOOT adjusts splits, offsets and cycle time from cyclic flow profiles measured upstream of the stop line, SCATS selects among plans using measured degree of saturation, and OPAC and RHODES solve a rolling-horizon optimisation over predicted arrivals. All four depend on a detector

plant installed and maintained in the carriageway, and all four were designed for streams in which lane discipline holds and vehicle classes are similar in size.

2.2 Max-pressure and backpressure control

Max-pressure signal control derives from backpressure scheduling in queueing networks [20]. Varaiya [18, 19] showed that repeatedly activating the phase maximising a weighted difference between upstream and downstream queues stabilises any demand within the capacity region using only local queue information, with no demand forecast and no coordination plan, which is attractive for a decentralised low-communication deployment. Related work established distributed formulations [21] and decentralised variants with weaker information requirements [22].

Two properties of the theoretical policy conflict with practice. First, the argmax is evaluated freely at every decision epoch, so the policy may switch as often as the decision interval allows, and since each switch costs a clearance interval during which no approach discharges, unconstrained switching wastes capacity; Levin et al. [23] address this with a cyclical phase structure, and our balanced-demand result in Section 6 is an independent, quantified instance of the same effect. Second, throughput optimality is proved for the unconstrained policy, and adding minimum green, maximum green, hysteresis and forced service breaks the argmax property and voids the proof. We therefore describe our controller as a queue-weighted pressure heuristic in the max-pressure family and make no stability claim.

2.3 Learning-based control

Reinforcement learning is the dominant research direction for signal control. Surveys by Wei et al. [24], Haydari and Yilmaz [25] and Noeen et al. [26] cover the field; representative works include IntelliLight [27], large-scale decentralised training [28] and PressLight [29], whose reward is derived from max-pressure and which is therefore a direct point of comparison. Early work by Genders and Razavi [30] established the basic formulation.

We did not adopt a learned policy, for reasons specific to the deployment context rather than to the merits of the methods: training requires a simulator calibrated to the target junction and none exists for the junctions we are concerned with; a neural policy does not naturally expose the maximum-wait or clearance guarantees an authority approving a signal plan needs; and it transfers poorly across geometries, whereas a pressure rule parameterised only by a phase set moves to a new junction by drawing polygons on a camera image. The comparison we do not make, and which should be made, is a head-to-head against a trained policy on the same arrival streams.

2.4 Camera-based counting and vision-driven signal control

Vehicle detection from fixed cameras is a mature application of single-stage detectors [9, 10], reviewed by Terven et al. [31], typically trained on general corpora [32] and extended with tracking when trajectories are required [33, 34]. Our system uses the Ultralytics implementation of the YOLO family [35]; the architecture generation we fine-tune is described in [36], which remains a preprint. Because the controller consumes instantaneous per-approach occupancy rather than cumulative crossings, we count detections inside approach polygons instead of tracking individual vehicles, which removes identity-switch errors in dense queues from the control path.

Recent applied studies pair such a detector with a signal timer. Papanashi et al. [14] report YOLO-based adaptive timing evaluated in Bengaluru, a heterogeneous-traffic context comparable to ours; Alruyshid et al. [15] add emergency and public-bus prioritisation in a Baghdad case study; Talaat et al. [16] and Medvei et al. [17] report detector-driven controllers, the latter combining detection with a learned policy. These works establish that the perception component is practical. Our contribution relative to them is on the control side: an explicit pressure formulation, stated

fairness bounds, a preemption latency bound, and an evaluation including a demand-optimal fixed-time baseline that reports where the adaptive policy loses. We are not aware of a published camera-based adaptive controller that reports a regime in which its own policy is worse than the fixed-time baseline; given the mechanism identified in Section 6.2 that regime almost certainly exists for those systems too, and its absence from the record is more likely a reporting bias than a property of the systems.

Emergency preemption has its own long operational history and a careful literature on its side effects, since a preemption interrupts the cycle and can create hazards at the transition [37]; Humagain et al. [38] review route optimisation and preemption methods, and recent work exploits connected-vehicle messaging to grant priority with explicit dilemma-zone protection [39]. Our approach differs in its detection channel, since the emergency vehicle is recognised visually by the same detector that counts ordinary traffic and no transponder or vehicle-side equipment is required. That is an advantage in cost and a disadvantage in reliability, which is why the ambulance class carries the highest confidence threshold and why the hold is capped, and what we add is a stated worst-case latency bound with a measurement against it.

3 System architecture

R.E.L.A.Y. consists of a perception stage and a control stage connected by a single data structure: a mapping from approach identifier to per-class vehicle count. Everything downstream of that mapping is independent of how the counts were obtained, which is what allows the same controller to be driven by recorded footage, a live camera, or a synthetic feed.

3.1 Perception

A single fixed camera views the junction. Each approach is defined by a polygon in normalised image coordinates, drawn once per site; no per-site model training is performed. For each frame a YOLO detector produces class-labelled boxes, which are filtered in four steps.

First, per-class confidence gating, because a single global threshold is unsuitable for mixed traffic: two-wheelers are small in the image and are lost at a threshold that suppresses false positives on cars, so the system uses 0.24 for motorcycles and bicycles, 0.38 by default, 0.50 for trucks and 0.68 for ambulances. The high ambulance gate is a safety choice, not an accuracy choice, since an ambulance detection moves the signal by itself and a marginal read on a white van would hand the junction to a non-existent emergency, while a vehicle below the gate still contributes ordinary demand through its next-best label. Second, cross-class deduplication, since per-class non-maximum suppression can leave one distant vehicle carrying stacked car, truck and motorcycle boxes: a greedy pass in descending confidence retains a box only if its intersection over union with every retained box is below 0.5, running after the confidence filters so a discarded low-confidence box cannot suppress a valid one. Third, approach assignment by centroid containment, giving one consistent rule for vehicles straddling a boundary, followed by an exponential moving average per approach and class with coefficient 0.35, roughly a one-second time constant at the observed detection rate. Fourth, failure handling: a bad frame returns the last smoothed counts rather than raising, and a staleness flag set after three seconds without a successful update makes the pipeline feed the controller empty counts, so the wait clocks and the maximum-wait rule cycle the phases safely instead of holding green on stale demand.

3.2 Control, topology and deployment

The controller consumes the count mapping and a set of conflict-free phases, and is topology-agnostic: four-arm, three-arm and two-arm junctions differ only in the phase set passed at construction, and approaches may be specified per arm or per movement, in which case the phase

Table 1: Passenger-car-unit weights w_c used in Equation (1). These are configuration values selected for a motorcycle-dominated mix. Their ordering is consistent with published equivalency estimates for heterogeneous South Asian traffic [40–42], but they were not estimated from local field measurements in this work; see Section 7.

Class	car	motorcycle	bus	truck	ambulance	bicycle	auto-rickshaw
w_c	1.0	0.3	2.5	2.5	2.0	0.2	0.8

set is generated to include paired through phases, paired protected-left phases and one split phase per arm serving both of that arm’s movements. Adding a junction therefore requires drawing polygons and declaring phases, not changing code.

Pedestrian demand enters as a separate transponder-style input giving, per arm, the number waiting, the longest wait and the number currently crossing. It is sanitised before use: non-numeric values, unknown arms, negative counts and non-finite waits are discarded, counts are clamped to 200 and waits to 180 s, so a client fault cannot inject an unbounded aging term.

The intended installation is one small computer per junction, taking a feed from an existing camera and driving the existing signal controller, with no central server in the control path and no junction-to-junction communication beyond the single downstream queue scalar the corridor configuration reads. We measure the perception cost in Section 6 but claim no cost figure, since no hardware was procured.

4 Control formulation

4.1 Notation and derived quantities

Let A be the set of approaches and P a set of phases, each $p \in P$ a subset of A that may be served simultaneously without conflict. Let Δt be the control step. At each step perception reports counts $n_{a,c}$ for approach a and class c , and a set of approaches on which an emergency vehicle was detected in the current frame.

Two distinct quantities are derived, the PCU-weighted demand and the occupancy indicator:

$$d_a = \sum_c w_c n_{a,c}, \quad (1)$$

with weights w_c from Table 1, and

$$o_a = \mathbb{1}[\sum_c n_{a,c} > 0]. \quad (2)$$

Keeping these separate matters: a single motorcycle contributes $d_a = 0.3$, which a demand threshold would treat as an empty approach, so the empty-phase skip and gap-out rules key on o_a rather than d_a and a lone two-wheeler still receives a full minimum green.

Each approach carries a clock r_a answering how long the oldest waiting vehicle on that approach has waited, rather than how long since the approach last had green. With G the set of approaches currently green and z_a counting consecutive seconds the approach has read empty,

$$r_a \leftarrow \begin{cases} 0 & a \in G, \\ r_a + \Delta t & a \notin G \text{ and } o_a = 1, \\ 0 & a \notin G, o_a = 0, z_a \geq \tau_e, \\ r_a & \text{otherwise,} \end{cases} \quad (3)$$

with $\tau_e = 3$ s. The third case is necessary rather than cosmetic: without it a long-vacant approach accumulates a large clock value, which the first arriving vehicle inherits, triggering forced service against a loaded corridor. The debounce keeps a detection dropout from erasing a real queue’s guarantee.

4.2 Phase score

Define a bounded urgency multiplier $u_a = 1 + \min(r_a, R)/R \in [1, 2]$, where R is the maximum-wait parameter. The vehicle pressure of a phase is

$$D(p) = \sum_{a \in p} d_a u_a. \quad (4)$$

Aging multiplies each approach's own demand and is capped at a factor of two, so waiting cannot manufacture demand that does not exist. This is a deliberate departure from an additive term $d_a + \lambda r_a$, which inverts the intended ordering: with $\lambda = 0.4$ a two-vehicle approach that has waited 45 s scores like an eighteen-PCU queue and the controller rotates phases at minimum-green cadence. For the same reason the multiplier is applied per approach rather than at phase level, since a phase-level $\max_a r_a$ lets a starving single-vehicle turn bay double the score of a phase whose large through queue has just been served.

Pedestrians are treated as demand. A phase p frees the crossing of arm x when neither x nor the opposing arm, whose straight-ahead movement would traverse the same crossing, is served: $F(p) = \{x : x \notin \text{arms}(p) \text{ and } \text{opp}(x) \notin \text{arms}(p)\}$. With m_x people waiting to cross arm x , k_x currently on that crossing and v_x the longest pedestrian wait there,

$$\Phi(p) = w_{\text{ped}} \left(\sum_{x \in F(p)} (m_x + k_x) \right) \left(1 + \frac{\min(\max_{x \in F(p), m_x > 0} v_x, V)}{V} \right), \quad (5)$$

with $w_{\text{ped}} = 0.3$, so one waiting person contributes about as much pressure as one motorcycle, and V the pedestrian maximum-wait parameter. Turning traffic across an open crossing yields to pedestrians in normal permissive operation, which is why $F(p)$ excludes only straight-ahead conflicts; counting pedestrians whose crossing is not actually freed causes the controller to hold greens that do not serve them. The complete score adds a large emergency constant $B(p) = B_0 \mathbb{I}[\exists a \in p : a \in \hat{E}]$ with $B_0 = 10^6$:

$$S(p) = D(p) + \Phi(p) + B(p). \quad (6)$$

A detection is a per-frame event and detectors drop frames, so each approach carries a hold timer reset to $\theta = 2.5$ s whenever an emergency is announced on that arm and otherwise decaying by Δt ; the latched set is $\hat{E}$. Announcements arrive at arm level and every movement key of that arm is latched, because the camera cannot resolve which lane the emergency vehicle occupies, so a single missed detection cannot abandon an active preemption.

4.3 Target selection and the stage machine

The signal cycles through green, yellow of duration y and all-red of duration ρ ; a green is never followed directly by a conflicting green. The next phase is chosen by strict precedence rather than by the score alone. (i) *Emergency*: if $\hat{E} \neq \emptyset$, take the phase serving the most emergency approaches, breaking ties by the largest r_a among them, so that on a movement-level junction the split-arm phase is chosen and whichever lane the emergency vehicle occupies turns green. (ii) *Vehicle starvation*: otherwise, if any approach has $r_a \geq R$ and $d_a > 0$, force a phase serving the worst such approach, preferring one that also serves approaches within 25 s of the threshold so that two clearance intervals are not paid where one would do; an empty approach cannot starve, so this never awards green to an empty lane. (iii) *Pedestrian starvation*: otherwise, if any arm has $m_x > 0$ and $v_x \geq V$, force a phase in $F(x)$ with the greatest pedestrian demand. (iv) *Score*: otherwise take $\arg \max_p S(p)$ over phases with $D(p) > 0$ or non-zero pedestrian demand, resting in all-red if no phase has demand.

A running green is terminated by the first applicable rule below, where e is elapsed green, $g_{\min}$ and $g_{\max}$ the minimum and maximum green, γ_t the gap-out time, Θ the preemption cap, W the walk-hold allowance and h the hysteresis margin.

Table 2: Controller parameters. The single-junction column is the configuration used in Experiments 1 and 3; the corridor column is the configuration of the system’s network demonstration, used in Experiment 2.

Symbol	Meaning	Single junction	Corridor
$g_{\min}$	minimum green (s)	5.0	5.0
$g_{\max}$	maximum green (s)	30.0	34.0
y	yellow (s)	3.0	2.0
ρ	all-red clearance (s)	1.5	2.5
γ_t	gap-out time (s)	2.5	2.5
R	vehicle maximum wait (s)	60.0	26.0
V	pedestrian maximum wait (s)	60.0	60.0
W	walk-hold allowance (s)	12.0	12.0
Θ	preemption cap (s)	60.0	25.0
θ	emergency latch (s)	2.5	2.5
h	hysteresis margin (PCU)	1.0	1.0
w_{ped}	pedestrian weight	0.3	0.3
γ	downstream coupling	n/a	0.4

1. An emergency exists on another phase and not the current one, and either $e \geq g_{\min}$ or the current phase is empty.
2. An emergency exists on the current phase: hold, subject to two bounds. An approach reaching $r_a \geq 2R$ forces service even over an active hold, and the cumulative hold is capped at Θ , after which the emergency term is dropped so the ordinary rules can move the signal. This prevents a stationary emergency vehicle from freezing the junction.
3. The current phase is empty, meaning no approach in p has $o_a = 1$ and no freed crossing has pedestrian demand, while another phase has demand: switch immediately without waiting for $g_{\min}$, since an empty phase has no vehicle to strand.
4. $e < g_{\min}$: hold. Minimum green is not overridden for a phase actually serving traffic.
5. Vehicle starvation elsewhere, then pedestrian starvation elsewhere: switch.
6. A pedestrian is still on a crossing freed by the current phase and $e < \min(g_{\max}, g_{\min} + W)$: hold. The bound matters, because an unbounded hold lets a renewing pedestrian stream pin green on a lightly used approach while a loaded corridor waits.
7. Gap-out: the served approaches have read empty for at least γ_t and another phase has demand.
8. Maximum green: $e \geq g_{\max}$, so green is yielded to the best other phase with demand.
9. Hysteresis: $S(p^*) > S(p) + h$ for the best alternative p^* , the margin suppressing oscillation between phases of nearly equal score.

Parameter values are listed in Table 2.

4.4 Corridor coupling and the preemption bound

On a corridor each junction j knows, for some arms, the downstream approach (j', a') that the arm’s discharge feeds. The demand presented to the score is then discounted by the downstream queue $q_{j', a'}$,

$$\tilde{d}_a = \begin{cases} \max(d_a - \gamma q_{j', a'}, \varepsilon) & o_a = 1, \\ 0 & o_a = 0, \end{cases} \quad \gamma = 0.4, \varepsilon = 0.05. \quad (7)$$

The floor at ε keeps the coupling a discount on pressure magnitude rather than an erasure of presence, since a physically occupied approach must remain occupied for Equations (2) and (3). Equation (7) is a one-sided approximation of the pressure difference used in max-pressure theory, where the difference is taken over turning movements with movement proportions; we use the aggregate downstream queue because it is the only downstream quantity one camera per junction can supply.

The worst-case delay from the first emergency detection to green on the emergency approach is bounded by the case in which the announcement arrives immediately after the controller has committed to a transition away from the very phase that serves the emergency. The committed transition completes, the newly served phase must be given its minimum green, and only then can the preemption take effect through a second clearance interval:

$$L_{\max} = y + \rho + g_{\min} + y + \rho. \quad (8)$$

With the single-junction parameters of Table 2, $L_{\max} = 14.0$ s.

The system also ships an executable self-check asserting twelve properties across four groups: clearance, green allocation (no green to an empty approach while another has demand, a full window for a lone motorcycle, and the $g_{\min}$ and $g_{\max}$ bounds), fairness (the R and V bounds and the walk hold), and emergency handling (prompt service, survival of a dropped detection frame mid-preemption, sequencing of emergencies on conflicting phases, and the Θ bound), plus input validation. All twelve pass on the version evaluated here and the captured output accompanies the experiment artefacts. These are executable assertions about behaviour under specified stimuli, not proofs, and they say nothing about the correctness of the perception input.

5 Experimental setup

5.1 Traffic model and baselines

Control results are obtained in a point-queue microsimulation, a standard abstraction in which delay accumulates at a single point rather than along a link [43]. While an approach displays green it discharges at $s = 0.55$ veh/s, a saturation headway of about 1.8 s, within the range capacity guidance gives for a through lane [44], accumulated as a fractional carry so a green shorter than one headway releases no vehicle, and with no banking of unused capacity while the queue is empty. Arrivals are Poisson per approach. We did not use an established microsimulator such as SUMO [45] because the controller under test is the system’s own code and the purpose is to drive that code on identical arrival streams; the cost is traffic-model fidelity, which we state as a limitation rather than defend.

The arrival process and the discharge model are the code shipped with the system under test, imported unchanged; the experiment scripts add warm-up handling, additional metrics and the baselines. Arrivals are drawn once per time step from a single seeded generator and pushed into every policy’s queue system, so the comparison is paired at the level of individual vehicles. Runs are 1800 s with the first 300 s discarded, because all queues start empty, and vehicles still queued at the end are counted with the wait accumulated so far, since discarding them would flatter whichever policy left the longer queue.

Two fixed-time baselines are used. The *equal-split* plan is a round-robin giving each phase 13.0 s of green, which with 3.0 s yellow and 1.5 s all-red produces a 35 s cycle; this is the baseline shipped with the system and represents the demand-blind plans the work aims to replace. The *Webster* plan takes cycle length and splits from Webster’s formulae [1] computed from the *true* mean demand of the run: with lost time $L = |P|(y + \rho)$ and critical flow ratios $y_i = \max_{a \in p_i} \lambda_a / s$ summing to Y , the cycle is $C = (1.5L + 5)/(1 - Y)$ clamped to $[30, 120]$ s and the effective green of phase i is $(C - L)y_i / Y$ floored at $g_{\min}$; when $Y \geq 0.95$ no finite Webster cycle exists and the cycle is set to the upper clamp with splits proportional to y_i . This baseline is deliberately generous

to fixed-time control, being granted perfect knowledge of the average demand and re-derived for every demand cell, and it is included because the equal-split baseline alone makes an adaptive controller look better than it is.

5.2 Experiments

Experiment 1: single junction. A four-arm junction with two phases, NS and EW, under three demand profiles. *Asymmetric* sets per-approach rates of (0.150, 0.130, 0.040, 0.035) veh/s for N, S, E and W, the shape the shipped benchmark samples at random, fixed here so a demand level is a stated number; *balanced* sets all four approaches to 0.089 veh/s; *single* loads one approach at 0.250 veh/s with 0.035 veh/s on the others. Each profile is scaled by multipliers $m \in \{0.6, 0.8, 1.0, 1.2, 1.4\}$, giving total demands of roughly 770 to 1790 veh/h, and each cell is run with 30 seeds. We report mean, median, 95th-percentile and maximum wait per vehicle, mean and maximum queue length and throughput; per-seed reductions are compared with a paired t -test and a Wilcoxon signed-rank test. A companion diagnostic (Experiment 1b) measures, over 10 seeds per cell, the phase change rate and the fraction of wall time during which some approach displays green; it explains Experiment 1 rather than adding a claim.

Experiment 2: three-junction corridor. A straight arterial carrying a three-arm junction, a four-arm junction and a second three-arm junction at 90 m spacing, with the topology, phase sets, routes and timing constants of the system’s network demonstration, and six routes: two arterial through movements and four entering from side arms. Link travel time is $90/12.5 = 7.2$ s at a free-flow speed of 12.5 m/s, so free-flow traversal of the three links is 21.6 s. Four policies are compared: the controller with coupling (Equation (7)) active, the same controller with $\gamma = 0$, an independent Webster plan per junction, and a coordinated Webster plan using a common cycle equal to the longest Webster cycle on the corridor with an eastbound progression offset of one link travel time per junction. Runs are 2400 s with 400 s warm-up, 20 seeds, at three demand levels, and one emergency vehicle is released from the west at $t = 1200$ s in every run, with other traffic yielding to it so that its traversal time isolates the signal contribution. The traffic layer is a re-implementation, because the system’s corridor demonstration is a browser application that cannot be driven headlessly; the control logic is the system’s own code.

Experiment 3: preemption latency. The junction is driven to a loaded state for a random interval between 40 and 160 s, after which an emergency vehicle is announced on a randomly chosen approach and re-announced every step thereafter. We record the delay until that approach displays green, with the stage and elapsed green at the instant of announcement, and replay the identical trial against the equal-split baseline, which has no preemption path. 300 trials are run at each of three demand levels.

Experiment 4: perception. The shipped detector is evaluated on the system’s held-out split of 97 images containing 4990 labelled instances across five classes. The full perception stack is then run over 60 s of two real fixed-camera junction clips at 960 px inference on an Apple M3 Pro using the Metal backend, recording per-frame latency and per-approach counts, with the controller running on those counts for the clip that shows two competing signalised streams so that the signal state in the exported figures is its real output. Finally, to test whether the counts have converged, the stack is run over frames sampled every 2 s from each clip at 640, 960, 1280 and 1600 px with temporal smoothing disabled.

6 Results

6.1 Single junction

Table 3 and Figure 1 give the single-junction results, in which three patterns should be separated.

Under asymmetric demand, the regime the system is designed for, R.E.L.A.Y. reduces mean

Table 3: Experiment 1. Mean delay per vehicle in seconds (standard deviation across 30 seeds in parentheses) and the paired per-seed reduction achieved by R.E.L.A.Y. against each baseline. Positive reductions favour R.E.L.A.Y.. All differences are significant at $p < 0.05$ by the Wilcoxon signed-rank test on paired per-seed means.

Profile	Demand (veh/h)	Mean delay per vehicle (s)			Reduction (%)	
		Equal split	Webster	R.E.L.A.Y.	vs equal	vs Webster
Asymmetric	767	11.0 (0.6)	8.4 (0.6)	5.1 (0.3)	+53.6 (4.0)	+38.8 (5.2)
	1022	11.8 (0.7)	8.9 (0.6)	5.9 (0.4)	+50.3 (4.9)	+34.1 (5.5)
	1278	13.9 (0.8)	9.6 (0.4)	7.0 (0.4)	+49.7 (4.9)	+27.0 (4.6)
	1534	19.2 (3.1)	10.7 (0.8)	8.3 (0.6)	+55.7 (7.2)	+22.0 (5.5)
	1789	45.1 (17.2)	11.8 (1.0)	10.1 (0.8)	+74.7 (8.4)	+14.2 (5.2)
Balanced	769	10.5 (0.6)	9.9 (0.4)	6.5 (0.4)	+38.1 (3.9)	+34.4 (3.5)
	1025	10.9 (0.4)	10.5 (0.4)	7.8 (0.5)	+28.7 (5.5)	+26.4 (4.9)
	1282	11.4 (0.3)	11.2 (0.4)	9.6 (0.7)	+16.1 (6.5)	+14.3 (5.9)
	1538	12.1 (0.6)	11.9 (0.6)	12.4 (1.3)	−2.7 (9.9)	−5.0 (10.4)
	1794	13.2 (0.6)	13.5 (0.6)	17.6 (2.0)	−33.0 (13.7)	−30.5 (13.3)
Single	767	14.4 (2.2)	8.8 (0.5)	5.5 (0.5)	+61.5 (4.2)	+37.6 (5.5)
	1022	38.3 (16.6)	9.8 (0.5)	6.9 (0.6)	+78.5 (8.8)	+29.4 (6.6)
	1278	151.8 (30.9)	11.8 (0.9)	9.1 (0.9)	+93.8 (1.1)	+22.9 (5.6)
	1534	239.8 (28.3)	17.1 (3.0)	13.1 (2.2)	+94.5 (0.9)	+22.1 (11.5)
	1789	307.2 (23.7)	21.3 (4.1)	28.3 (10.3)	+90.8 (2.9)	−33.0 (36.4)

delay by 49.7% against the equal-split plan and by 27.0% against the Webster plan at 1278 veh/h. The gain against the equal-split plan grows with demand, reaching 74.7% at 1789 veh/h, but that growth is a capacity artefact rather than a control improvement: at that demand the equal-split plan is oversaturated on the major approaches and its mean delay diverges, whereas the Webster plan and R.E.L.A.Y. both remain stable at essentially the same throughput, 1804 and 1800 veh/h. The gain against the Webster plan moves the other way, falling monotonically from 38.8% to 14.2%, because a well-timed fixed plan converges toward the best achievable allocation as saturation approaches. We regard the Webster column as the honest headline.

Under a single dominant approach the equal-split plan collapses: at 1278 veh/h its mean delay is 151.8 s, its mean queue is 55.4 vehicles and its throughput falls to 1123 veh/h against a demand of 1278 veh/h, so it is not serving the traffic offered. Against the Webster plan the gain is a much more modest 22.9%, and at the highest demand the sign reverses to −33.0% with a standard deviation of 36.4 percentage points. The mechanism is identifiable: the Webster plan for that cell allocates a 51.2 s green to the loaded phase within a 67.4 s cycle, whereas $g_{\max} = 30$ s caps R.E.L.A.Y.’s green, so the adaptive controller pays clearance more often on an approach whose demand is close to its capacity. That is a parameter choice showing through, not a property of the pressure rule, and it argues for making $g_{\max}$ demand-dependent.

Under balanced demand the advantage decays and becomes a penalty: +34.4% at 769 veh/h, +14.3% at 1282 veh/h and −30.5% at 1794 veh/h against the Webster plan. This is a real regression, whose cause Section 6.2 identifies.

Figure 2 shows the wait distribution rather than the mean, which is where the fairness bound becomes visible. At 1278 veh/h under asymmetric demand the 95th-percentile wait is 29.9 s under the equal-split plan, 25.8 s under the Webster plan and 15.8 s under R.E.L.A.Y., with worst observed waits averaging 50.0 s, 50.1 s and 29.6 s. Under a single dominant approach the contrast is larger: 361.5 s and 72.9 s against 40.5 s under R.E.L.A.Y., consistent with the forced-service rule at $R = 60$ s. Figure 3 shows the same comparison unaggregated, one point per seed, which makes both the consistency of the gain and the crossover in the balanced profile directly visible.

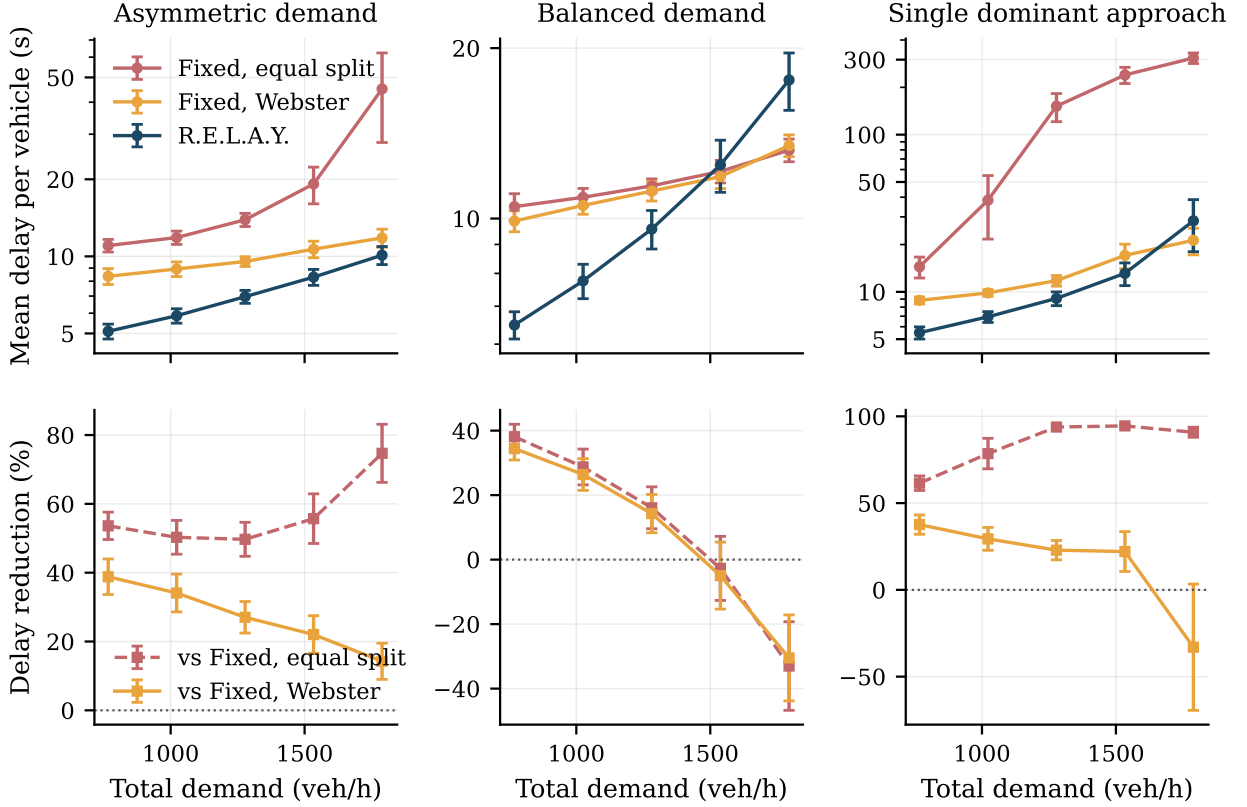


Figure 1: Experiment 1. Top row: mean delay per vehicle against total demand for the three demand profiles, on a logarithmic axis. Bottom row: paired per-seed delay reduction achieved by R.E.L.A.Y. against each fixed-time baseline. Error bars are one standard deviation across 30 seeds. The balanced profile crosses zero near 1500 veh/h.

6.2 Why the balanced case regresses

Figure 4 explains the regression, and the explanation is a measurement rather than an argument. Under balanced demand the controller has no basis for preferring one phase over the other, so successive scores are close, the hysteresis margin of 1.0 PCU is crossed frequently, and green terminates near the minimum. Mean green duration under balanced demand is only 7.3 s at the lowest level and falls to 5.3 s at $m = 1.2$, barely above $g_{\min} = 5$ s, against 12.0 s under asymmetric demand at the same lowest level, and the phase change rate rises from 306 to 364 per hour against 204 per hour for the equal-split plan. Because every change costs $\gamma + \rho = 4.5$ s of clearance, the fraction of time during which some approach displays green falls from 0.743 under the fixed plan to 0.543 under R.E.L.A.Y., a loss of 20 percentage points of usable capacity. At low demand that loss is affordable because there is spare capacity; at high balanced demand it is not, and it exceeds the benefit of reallocating green.

The asymmetric and single profiles do not show this to the same degree, because there the score gap between phases is large and stable, so the incumbent keeps winning and greens stay long: mean green under the single-approach profile at $m = 1.4$ is 14.0 s, longer than the fixed plan’s 13.0 s, and the switch rate of 193 per hour is below the fixed plan’s 204.

This is the practical form of a known theoretical issue. Max-pressure control as formulated in the literature evaluates its argmax freely and imposes no cycle structure, and Levin et al. [23] propose a cyclical phase structure precisely to recover the lost capacity. Our data quantify what the unconstrained version costs in an easily reached regime, and indicate that any deployment should include a saturation-aware guard that lengthens minimum green or falls back to a cyclical structure when the measured degree of saturation is high and phase scores are close. That guard is not implemented in the version evaluated here, and implementing and evaluating it is the most

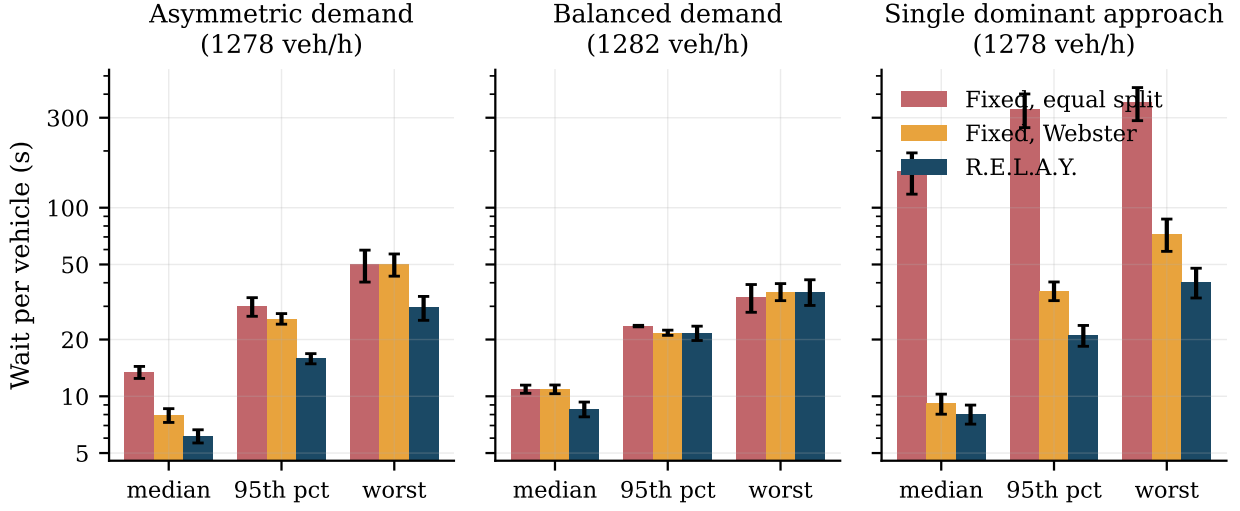


Figure 2: Experiment 1. Median, 95th-percentile and worst observed wait per vehicle at demand multiplier 1.0, on a logarithmic axis. The tail, not the mean, is where the anti-starvation bound shows: under a single dominant approach the equal-split plan’s worst wait averages 361.5 s against 40.5 s for R.E.L.A.Y..

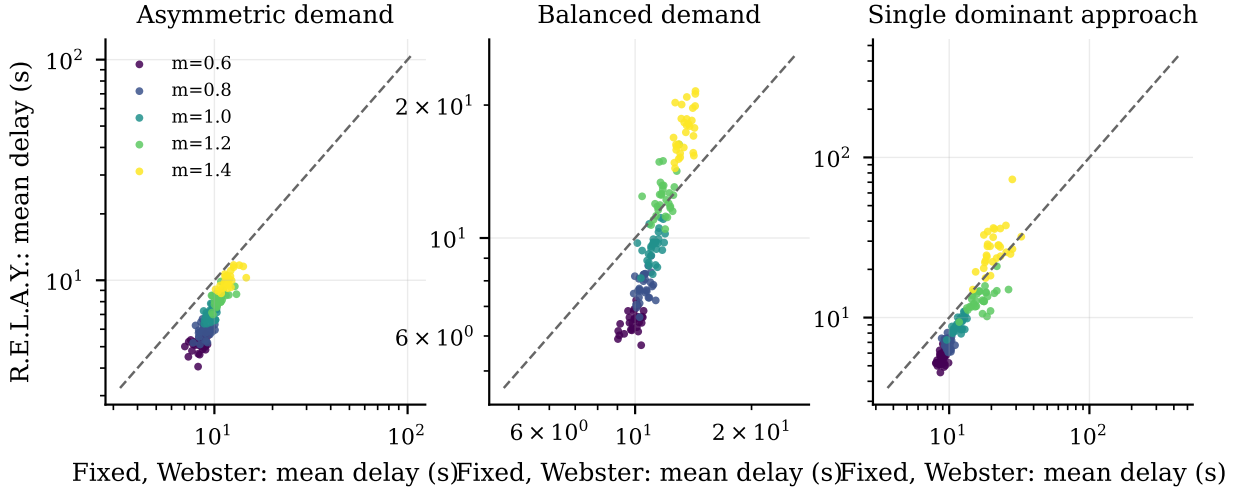


Figure 3: Experiment 1, unaggregated. Each point is one seed at one demand level, plotting R.E.L.A.Y.’s mean delay against the Webster plan’s mean delay on the identical arrival stream. Points below the diagonal favour R.E.L.A.Y.. Colour encodes the demand multiplier from 0.6 (dark) to 1.4 (light).

valuable single piece of future work these data indicate.

6.3 Corridor

Table 4 and Figure 5 contain three findings, one of them negative.

First, the delay reduction is large at low and moderate demand and small at high demand: 57.3% and 44.9% against the independent Webster plans at 769 and 1282 veh/h, falling to 11.2% at 1794 veh/h, with 55.6%, 43.1% and 7.3% against the coordinated plan. The coordinated plan is only 2 to 4% better than the independent plans on mean delay, as expected, since progression offsets improve the through movement and not the average over all movements.

Second, the coupling term does essentially nothing. Disabling it, which turns the corridor into three independently adaptive junctions, changes mean delay by -0.2% , -0.4% and -2.5% at the three demand levels, and at the highest demand the uncoupled variant is marginally better. The corridor benefit is therefore attributable to local adaptivity, not to coordination. We report this plainly because the coupling term is part of the system’s design rationale and the data do

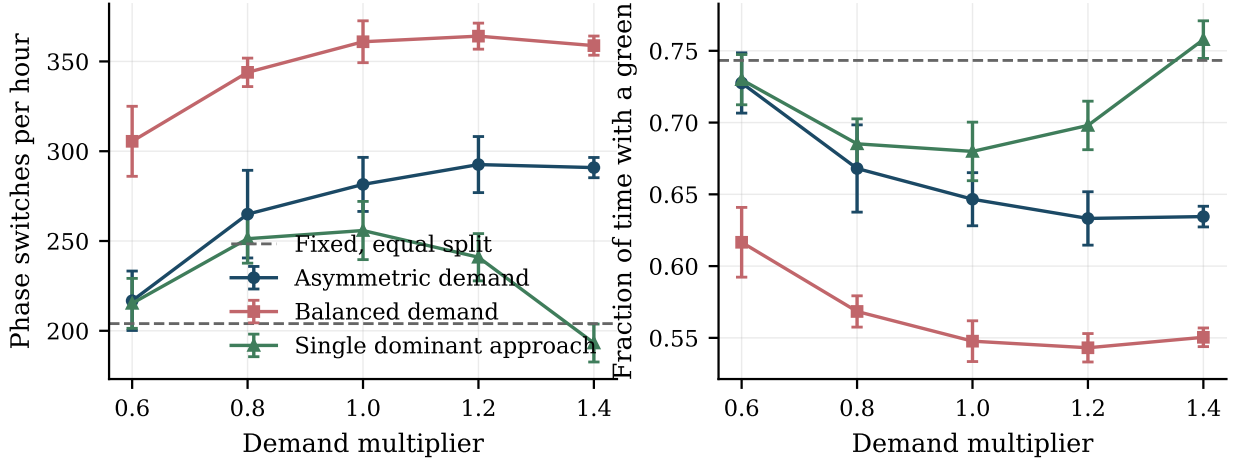


Figure 4: Experiment 1b. Left: phase changes per hour. Right: the fraction of wall time during which a green is displayed, the complement of which is time spent in yellow and all-red clearance. The dashed line is the equal-split fixed plan, whose behaviour is independent of demand by construction.

not support it. Two explanations are consistent with our setup and we cannot separate them here: the point-queue model has no spatial extent, so a downstream queue never physically blocks an upstream discharge, which removes the main mechanism the coupling addresses; and a single aggregate downstream queue is a weak proxy for the movement-specific pressure difference max-pressure theory uses.

Third, and less favourably, at the highest demand R.E.L.A.Y. attains lower mean delay than either fixed plan, 32.6 s against 36.7 s and 35.3 s, but *worse* arterial through travel time, 46.9 s against 43.1 s and 39.3 s. The controller minimises total delay across all movements and at saturation does so partly by serving side arms a coordinated plan would make wait, so a city whose objective is arterial progression rather than total delay would prefer the coordinated fixed plan in that regime. This is a genuine objective mismatch, not a bug, and it should be stated when such a system is proposed to an operator.

The emergency result is unambiguous. The emergency vehicle traverses the corridor in 22.2 to 26.1 s under R.E.L.A.Y. against a free-flow reference of 21.6 s, so it is delayed by roughly 1 to 5 s across three signalised junctions, against 30.1 to 53.6 s under the independent Webster plans. The coordinated plan does well here, 24.0 to 31.5 s, because the emergency vehicle travels in the progression direction; a westbound emergency would not benefit.

6.4 Emergency preemption latency

Mean latency is 4.19 s against 7.00 s for the fixed cycle, and is almost insensitive to demand, rising only from 3.99 s to 4.47 s between the lightest and heaviest loads (Table 5). The important quantity is the tail: the 95th percentile is 11.28 s against 20.59 s and the worst case over 900 trials is 13.85 s against 22.29 s. The measured worst case sits 0.15 s below the analytical bound of 14.0 s from Equation (8), and inspecting the worst trials confirms the predicted mechanism, since every trial above 13.4 s is one in which the announcement arrived within the first second of a yellow already transitioning away from the emergency’s own phase. Because the bound is composed only of clearance and minimum-green settings, it can be tightened by configuration and stated in advance to an emergency service, which a demand-blind fixed plan cannot do.

Two qualifications belong with this result. In 294 of 900 trials, or 32.7%, the emergency approach already had green when the vehicle was detected, so latency was effectively zero and the mean is pulled down accordingly; and in 309 trials, or 34.3%, the fixed cycle happened to reach that approach sooner than the preemption did, which is why the distributions overlap substantially in Figure 6 even though tail behaviour differs sharply. The claim these data support is a bounded

Table 4: Experiment 2. Three-junction corridor, mean over 20 seeds with standard deviation in parentheses. Delay is total queueing delay per vehicle across all junctions traversed. Through travel time is measured for the two arterial through routes only; free-flow traversal of the three links is 21.6 s. Emergency traversal is the single emergency vehicle released in each run, whose standard deviation is zero for the fixed plans because those plans are deterministic.

Demand (veh/h)	Policy	Mean delay per vehicle (s)	Through travel time (s)	Stops per vehicle	Emergency traversal (s)
769	Webster, independent	26.7 (0.9)	39.7 (0.7)	1.95	36.0 (0.0)
	Webster, coordinated	25.7 (0.7)	37.0 (0.6)	1.78	24.5 (0.0)
	R.E.L.A.Y., no coupling	11.4 (0.5)	25.0 (0.5)	1.10	22.2 (3.4)
	R.E.L.A.Y., coupled	11.4 (0.5)	25.1 (0.5)	1.10	22.2 (3.4)
1282	Webster, independent	30.1 (0.8)	41.5 (0.6)	2.09	53.6 (7.5)
	Webster, coordinated	29.2 (1.1)	38.0 (0.7)	1.80	31.5 (0.0)
	R.E.L.A.Y., no coupling	16.5 (0.9)	29.8 (0.9)	1.58	23.4 (4.6)
	R.E.L.A.Y., coupled	16.6 (0.8)	30.1 (0.8)	1.59	23.8 (4.4)
1794	Webster, independent	36.7 (3.3)	43.1 (1.0)	2.10	30.1 (6.5)
	Webster, coordinated	35.3 (3.6)	39.3 (1.1)	2.03	24.0 (0.0)
	R.E.L.A.Y., no coupling	31.9 (4.2)	45.4 (4.4)	2.08	26.2 (8.3)
	R.E.L.A.Y., coupled	32.6 (4.0)	46.9 (4.4)	2.10	26.1 (7.9)

Table 5: Experiment 3. Delay from the first emergency detection to green on the emergency approach, over 900 trials. The equal-split baseline has no preemption path, so its latency is determined by cycle position and is independent of demand, which is why its statistics are pooled.

Demand	Policy	Mean (s)	SD (s)	Median (s)	95th pct (s)	Worst (s)
Light ($m=0.6$)	R.E.L.A.Y.	3.99	3.55	4.58	10.77	13.61
Medium ($m=1.0$)	R.E.L.A.Y.	4.11	3.78	4.51	11.26	13.80
Heavy ($m=1.4$)	R.E.L.A.Y.	4.47	3.94	4.61	12.10	13.85
All	R.E.L.A.Y.	4.19	3.76	4.56	11.28	13.85
All	Equal split	7.00	7.04	5.02	20.59	22.29

worst case, not a uniformly faster response.

6.5 Perception

Table 6 reports mean average precision of 0.954 at IoU 0.5 and 0.809 averaged over IoU 0.5 to 0.95, with precision 0.950 and recall 0.886. Per-class average precision exceeds 0.96 for cars, buses, trucks and ambulances and is lowest for motorcycles at 0.851, the expected ordering given that two-wheelers are the smallest and most frequently occluded class. An important caveat concerns what the split is: it mixes automatically labelled frames from a synthetic feed with frames from real footage pseudo-labelled by a general detector rather than by human annotation, so a high score establishes that the fine-tuned model reproduces the labelling function it was trained against, not accuracy against human ground truth, and the ambulance figure of 0.987 reflects a class that is easy to separate in the synthetic domain. We deliberately omit a comparison against general COCO-trained checkpoints on this split, because their class indices do not correspond to this label space and any such comparison would be an artefact of index mismatch.

Running the full perception stack on real footage at 960 px inference on an Apple M3 Pro, median per-frame latency is 38.3 ms on the boulevard clip (26.1 frames per second, 95th percentile 85.7 ms) and 85.8 ms on the dense mixed-traffic clip (11.7 frames per second, 95th percentile 130.7 ms); the first frame of a run costs 1735 ms because of one-off backend initialisation and is excluded. Both rates exceed what the control loop needs, since decision epochs are governed by the 5 s minimum green rather than by the frame rate, and perception is decoupled from control

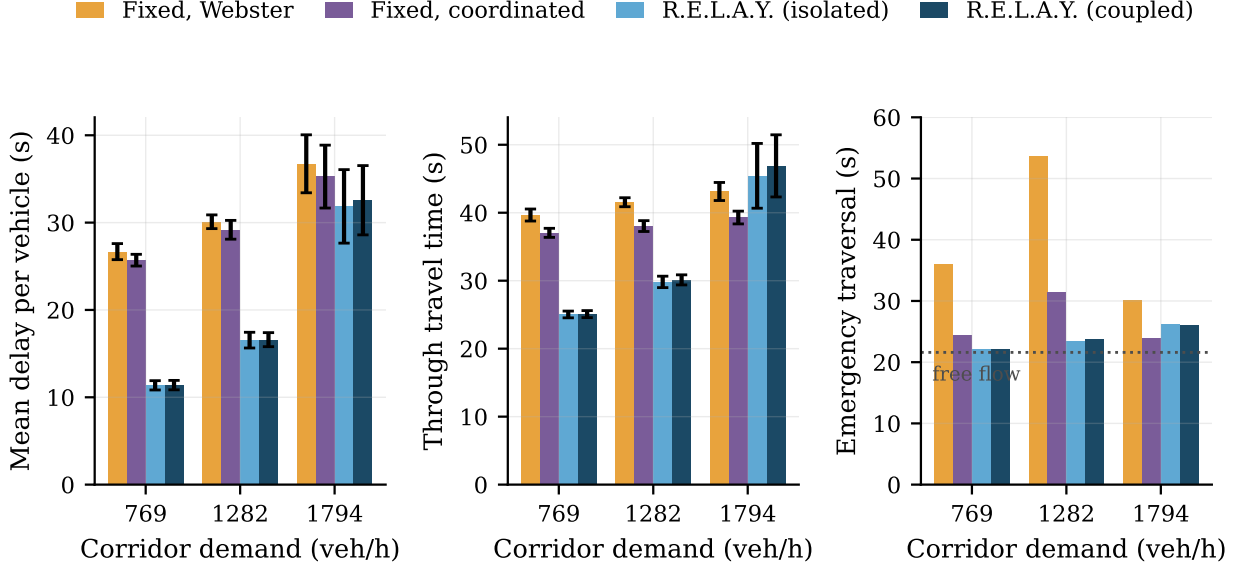


Figure 5: Experiment 2. Left: mean queueing delay per vehicle across the corridor. Centre: travel time for the two arterial through routes; at the highest demand the coordinated fixed plan gives through traffic a better ride than either adaptive variant. Right: traversal time of the single emergency vehicle released in each run, against the 21.6 s free-flow reference.

Table 6: Experiment 4. Detector performance on the system’s held-out split: 97 images, 4990 labelled instances. The checkpoint evaluated is the one shipped with the system, a 2.59-million parameter model. P and R are precision and recall at the operating point the validator reports.

Model	Overall				AP@50 by class				
	mAP@50	mAP@50-95	P	R	car	motorcycle	bus	truck	ambulance
Shipped fine-tune	0.954	0.809	0.950	0.886	0.985	0.851	0.981	0.964	0.987

so a latency spike delays a count update rather than blocking a decision.

Figure 8 shows annotated frames from both clips, and the second shows the main perception weakness plainly. To quantify rather than assert it, we measured how the count changes with inference resolution (Figure 7, right). On the dense clip the mean sampled vehicle count rises from 21.4 at 640 px to 26.5 at 960 px, 33.2 at 1280 px and 39.1 at 1600 px, an increase of 83% with no sign of convergence, and motorcycles alone rise by 120%; on the sparser boulevard clip the count rises from 4.85 to 10.0. A count that keeps growing with resolution has not converged, so the absolute demand values the controller receives on dense two-wheeler scenes are underestimates of unknown magnitude. Median inference cost over the same range rises from 47.2 ms to 108.0 ms, so raising resolution is not free.

This weakness does not immediately invalidate the control results, because the controller compares approaches rather than reading absolute volumes, so a systematic undercount affecting both approaches similarly leaves the ordering of phase scores intact. What it does affect is any rule keyed to an absolute quantity, in particular the occupancy indicator o_a near an empty approach and the gap-out timer. A regionally fine-tuned detector on human-annotated South Asian footage is the direct remedy and is not attempted here.

Taken together, these results support a narrower claim than the one usually made for camera-based adaptive control. The controller helps most where demand is imbalanced and the junction is not saturated, and its operational value is clearest in the bounds rather than the means: the worst observed wait tracks the configured R across every profile and the preemption worst case is bounded in closed form, so an authority can be told in advance what the guarantees are. The coordination term is not supported by our data, and the pedestrian logic, though verified by

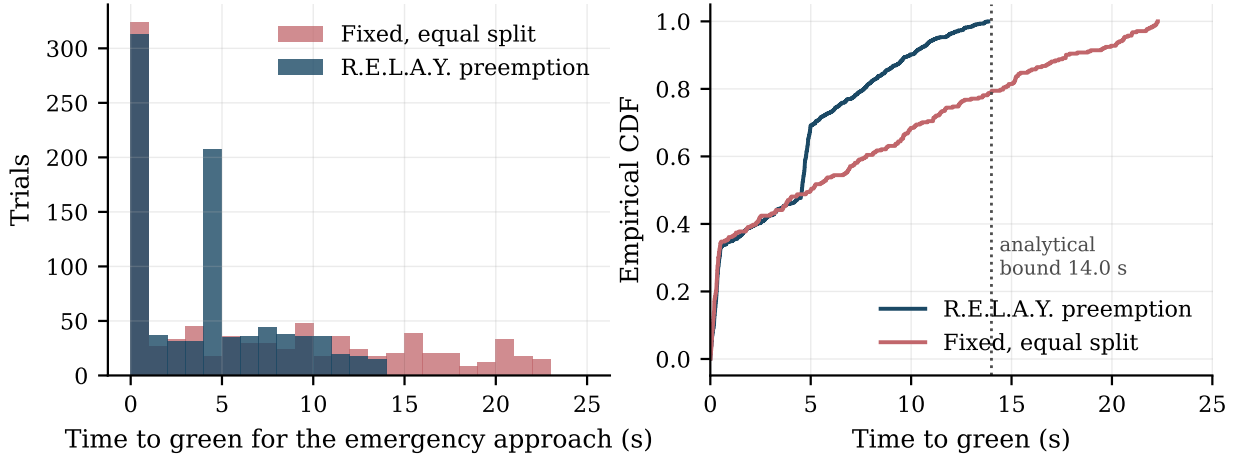


Figure 6: Experiment 3. Left: histogram of the time from the first emergency detection to green on the emergency approach, over 900 trials. Right: the same data as empirical cumulative distributions against the analytical bound of Equation (8).

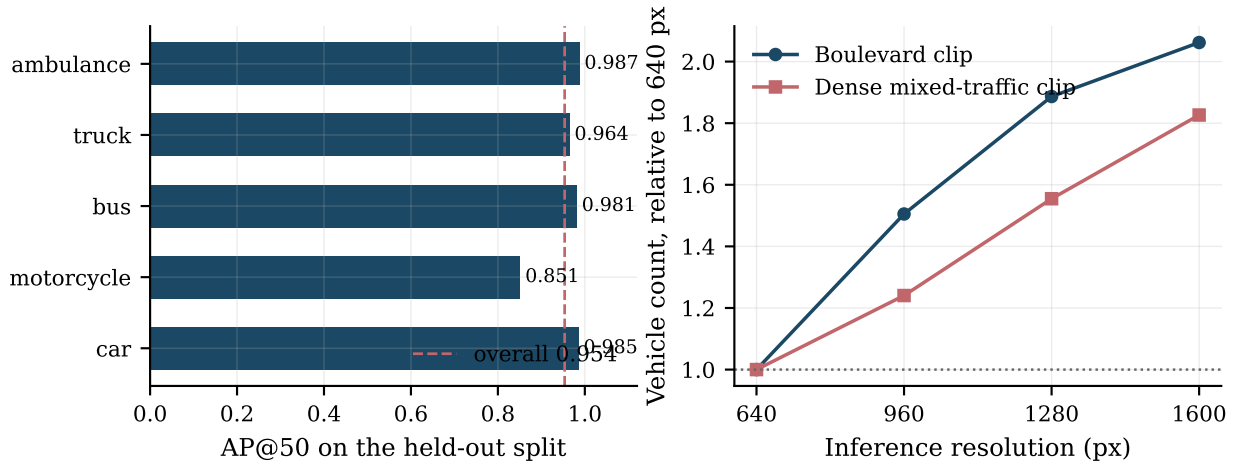


Figure 7: Experiment 4. Left: per-class average precision at IoU 0.5 on the held-out split, with the overall mean average precision marked. Right: sampled vehicle count as a function of inference resolution, normalised to the count at 640 px. Neither curve has flattened by 1600 px, which is direct evidence that vehicles are still being missed.

executable invariants, was never benchmarked against a pedestrian-aware baseline, so we claim no pedestrian delay benefit.

7 Limitations

We list the limitations we believe would most affect a reader’s assessment, in descending order of importance.

This is not a field study. No signal was controlled. There is no controller cabinet in the loop, no communication with a real signal and no before-and-after measurement, so we have no evidence about the failure modes that appear only in the field: power interruption, lens contamination, camera repositioning, seasonal light changes, or the behaviour of drivers who learn that the signal responds to them. Every quantitative claim should be read with that scope.

The traffic model is a point queue. Queues have no spatial extent, so there is no spillback, no blocking of an upstream junction by a downstream queue, no lane changing and no capacity loss from vehicles obstructing the junction box. The abstraction is favourable to any adaptive policy, whose mistakes are cheaper when queues cannot physically block anything, and specifically

unfavourable to our coordination term, whose purpose is to prevent a jam being fed, so the corridor coupling result should be read as inconclusive rather than negative and a spatially explicit model in SUMO [45] would settle it.

Perception was neither validated against human ground truth nor propagated into the control experiments. The held-out split mixes synthetic frames with pseudo-labelled real frames, so the reported mean average precision of 0.954 measures agreement with the labelling process rather than accuracy against human annotation; we can state that counts on dense footage are underestimates that have not converged by 1600 px inference, but not the error magnitude, and a human-annotated evaluation set from the target city is required before any absolute demand claim. Separately, the control experiments feed the controller exact queue counts from the simulation, whereas the real system feeds it smoothed detector counts with those biases, so we have not measured how much of the delay reduction survives realistic count error. This is arguably the most important open question for the design. The controller contains mitigations (temporal smoothing, presence separated from demand, and a staleness fallback that cycles the phases when perception stops producing counts), but mitigations in the code are not measured robustness, and a sensitivity study injecting calibrated count noise and dropout into Experiment 1 should be done.

The PCU weights are configuration, not calibration. The values in Table 1 were chosen to reflect a motorcycle-dominated mix rather than estimated from local field measurements. Because they directly scale the pressure in Equation (4), a mis-specified motorcycle weight biases every phase comparison at a junction whose approaches differ in class mixture, and we ran no sensitivity analysis over them.

Four narrower gaps. The single-junction experiments use a two-phase junction, so although the controller supports movement-level phasing and generates protected-left phases, the results do not exercise it; a junction with five or six phases has proportionally more lost time per cycle, which the balanced-demand result suggests would amplify rather than reduce the switching-cost problem. We compare only against fixed-time plans, whereas the natural research baseline for a pressure-based controller is a learned policy, PressLight [29] in particular since its reward derives from max-pressure; our reasons for not deploying one are practical and stated in Section 2, but they do not excuse the absence of a comparison that would locate this controller between fixed-time and learned control. The pedestrian logic is asserted by executable invariants but never compared against an alternative treatment such as a conventional fixed pedestrian stage, so we report its behaviour and not its benefit. Finally, the values in Table 2 came from iterative development against the system’s own demonstrations rather than from optimisation over the evaluation scenarios, which cuts both ways: the results are not the product of tuning on the test scenarios, and the controller is probably not at its best, as the interaction between $g_{\max}$ and the single-approach result at high demand shows.

8 Conclusion

We have described a single-camera adaptive signal controller for heterogeneous urban traffic and evaluated it in a seeded microsimulation and on real fixed-camera footage. The controller converts per-approach, per-class detections into PCU-weighted demand and selects phases by a queue-weighted pressure score in the max-pressure family, with the operational rules a deployable signal requires: clearance intervals, minimum and maximum green, empty-phase skipping, gap-out, a hard maximum-wait bound for vehicles and a separate one for pedestrians, and emergency preemption with a bounded hold.

Measured against a fixed-time plan given the true mean demand, the controller reduces mean delay by 14 to 39% wherever demand is imbalanced, cuts the 95th-percentile and worst-case waits substantially further than it cuts the mean, and brings an emergency vehicle to green within a worst case of 13.85 s against an analytical bound of 14.0 s.

Three results qualify those figures and are reported rather than suppressed. Under balanced saturated demand the controller is 30.5% worse than the Webster plan, because unconstrained switching raises the phase change rate by 76% and reduces displayed green time by 20 percentage points, so a cyclical phase structure or a saturation-aware guard is required before deployment. The corridor coordination term produces no measurable benefit in our model. And at high corridor demand the controller achieves lower total delay but worse arterial through travel time than a coordinated fixed plan, an objective mismatch an operator should be told about. The next steps follow from the limitations: a spatially explicit corridor model to settle whether the coordination term is useful, a count-noise sensitivity study, a human-annotated evaluation set from the target city, and a saturation-aware switching guard evaluated in the balanced regime.

Data and code availability

The controller, perception stack and simulation harness are available at <https://github.com/kafle1/relay> under the AGPL-3.0 licence. The experiment scripts that produced every table and figure, the raw per-run result files, the captured invariant-check output and the figure generation script are distributed with this manuscript in the `experiments/` directory. Re-running the experiment scripts and then the figure script reproduces every number and every panel reported here.

Declarations

The author declares no competing interests and received no funding for this work. No human subjects or personal data were involved: the footage used is publicly available recorded video of public roads, used only for aggregate vehicle counting, and no identification of individuals or vehicles was performed or attempted.

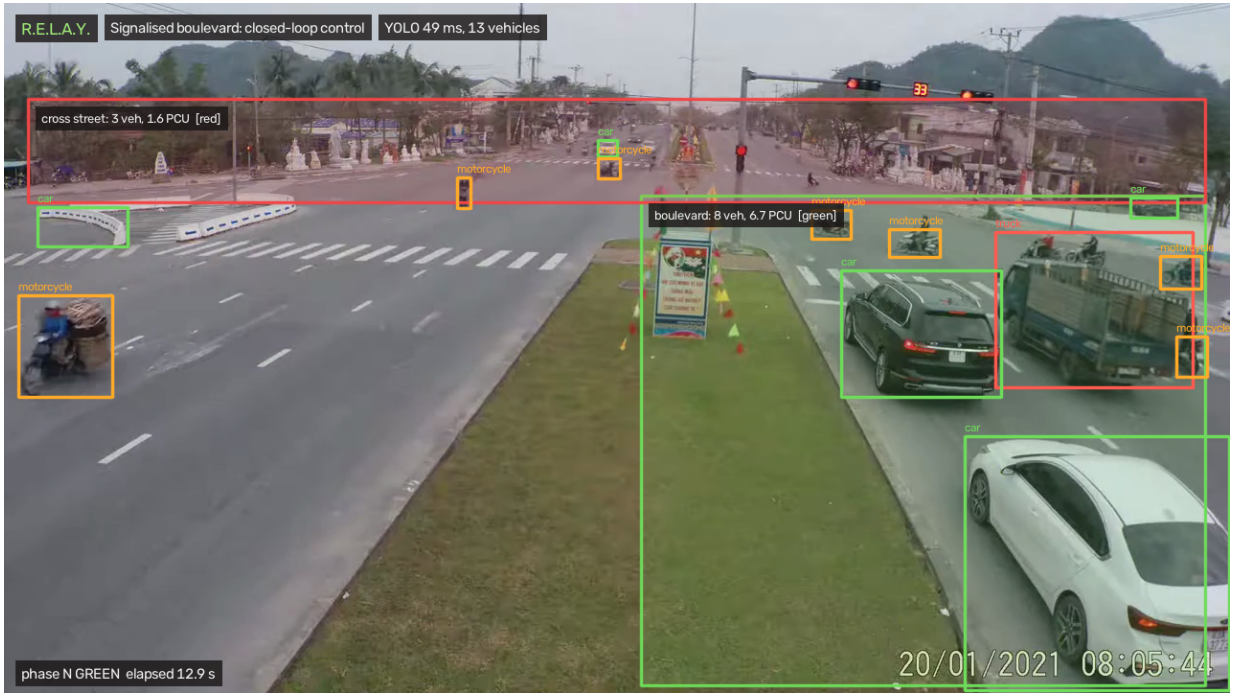
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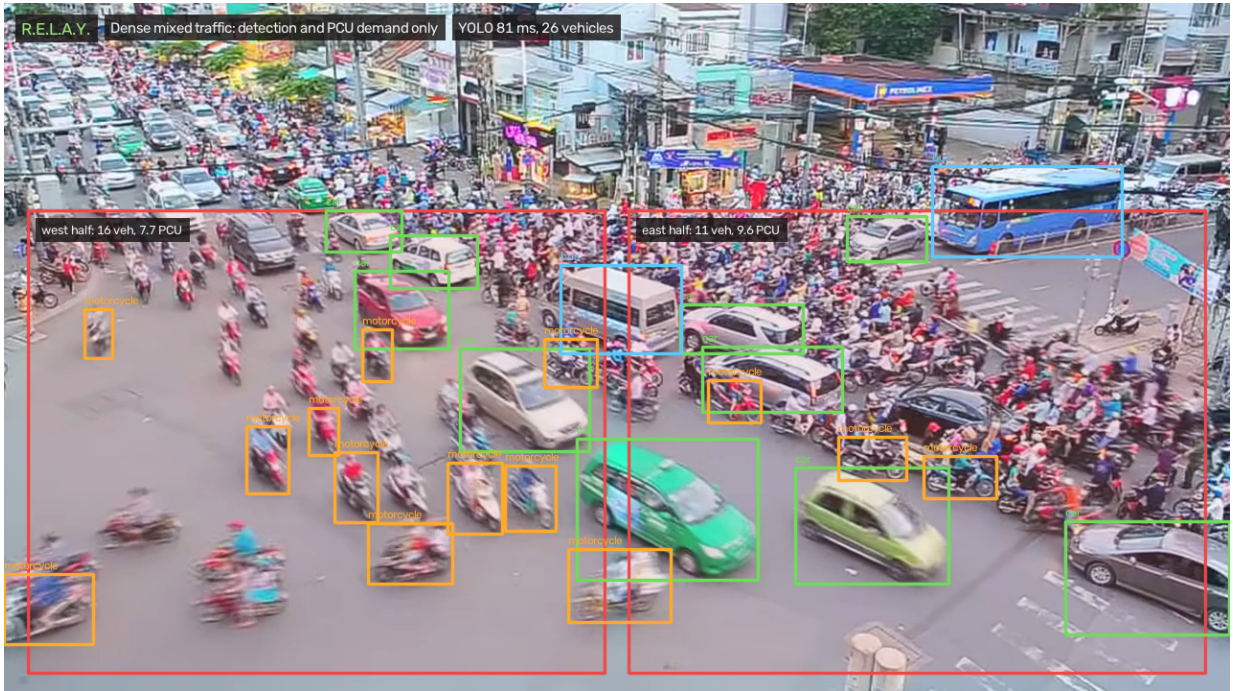
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(a) Signalised boulevard. The controller is running on these counts: the near boulevard approach holds 8 vehicles and 6.7 PCU and has green, the crossing street holds 3 vehicles and 1.6 PCU and is red, and the header reports the real inference latency for the frame.



(b) Dense motorcycle-dominated traffic. The failure mode is visible: 26 vehicles are detected in a scene containing several hundred, so the reported PCU demand of 7.7 and 9.6 for the two halves is a severe underestimate. No controller decision is shown, because the junction depicted is not signal-controlled in the way the two zones imply.

Figure 8: Experiment 4. The perception stack on real fixed-camera footage. Both frames are direct output of the system, with detections, approach polygons and per-approach PCU demand as computed at run time.